Notes
Biostatistical Methods II

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0.1 Introduction

The notes in this PDF are based on the course "Biostatistical Methods II" taught by Dr. Datta at UF in the Fall semester of 2023. The course, and these subsequent notes, do not follow any textbooks and are solely made from the notes written by Dr. Datta; who has also provided a list of books in the course syllabus which can be easily found online. In addition, coding examples are provided and will be integrated into these notes where applicable.
A word of caution, these are notes and not a textbook. These are under the edit of no professor and should be considered as trustworthy as a blog post. It should be noted that if you do not confirm the equations yourself (which you should for practice) then the notes cannot be trusted.

0.1.1 Overview

The notes will cover linear models from distributions within the exponential family, including the theory, estimation, and testing procedures for multiple models. Full mastery of the topics cannot be gained from studying these notes, but some naive examples and proofs will be elaborated upon. No home works will be covered in these notes.

1 The General Linear Model

Ordinary normal linear regression models are written as

$$y = X\beta + \epsilon,\ \epsilon \sim N(0, \sigma^2),\ \text{where } E(y_i) = \mu_i = X^T B,\ \ \text{which implies}\ \ y_i \sim N(\mu_i, \sigma^2)$$

An ordinary linear regression model is a generalization of a generalized linear regression model (GLM). GLMs allow the linear predictors to be related to the response variable via a continuous functions rather than directly. It also allows the magnitude of the variance of each measurement to be a function of its predicted value. The GLM consists of three elements:

1. A particular distribution for modeling Y from among those which are considered exponential families of probability distributions.

2. A linear predictor $\eta = X\beta$.

3. A *link function*, g, such that $E(Y|X) = \mu = g^{-1}(\eta)$.

1.1 The Link Function

The link function may be simple or complex, but must always be differentiable (i.e. $g(\mu_i) = X_i^T\beta$, or $g(\mu_i) = \log(\mu_i) = X_i^T\beta$). When $g(\mu_i) = \mu_i = X_i'\beta$, then this is the *identity link*.
The link function is the heart of GLMs and each link function discussed comes with different interpretations. This is the crux of the course.

2 The Exponential Family

In a GLM, each outcome Y of the dependent variables is assumed to be generated from a particular distribution in an exponential family, a large class of probability distributions that includes the

normal, binomial, Poisson and gamma distributions, among others. The exponential family distribution's can all be written similarly. Let $f(y;\theta,\phi)$ be in the exponential family, then $f(y;\theta,\phi)$ can be written as follows:

$$f(y;\theta,\phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right) \tag{1}$$

where ϕ is a vector of *known parameters*, often called *nuisance parameters* and $b(\theta)$ is monotonic. (1) is referred to as the *"Natural Form"*, or the *"Canonical Form"*, of the exponential family. The function $a_i(\phi)$ has the general form

$$a_i(\phi) = \frac{\phi_i}{p_i}$$

where p_i is a known prior, in most cases $p_i = 1$.

2.1 common distributions in exp family form

2.1.1 Binomial(n,π)

$$\begin{aligned}
f(y;\pi) &= \binom{n}{y}\pi^y(1-\pi)^{n-y} \\
&= \exp\left(\log\binom{n}{y} + y\log(\pi) + n\log(1-\pi) - y\log(1-\pi)\right) \\
&= \exp\left(y\log\left(\frac{\pi}{1-\pi}\right) - (-n\log(1-\pi)) + \log\binom{n}{y}\right)
\end{aligned}$$

where

$$\begin{aligned}
a(\phi) &= 1 \\
\theta &= \log\left(\frac{\pi}{1-\pi}\right) \\
b(\theta) &= -n\log(1-\pi) \\
c(y,\phi) &= \log\binom{n}{y}
\end{aligned}$$

2.1.2 Exponential(μ)

$$\begin{aligned}
f(y;\pi) &= \mu\exp(-\mu y) \\
&= \exp\left(y(-\mu) - (-\log(\mu))\right)
\end{aligned}$$

where

$$\begin{aligned}
a(\phi) &= 1 \\
\theta &= -\mu \\
b(\theta) &= -\log(\mu) = -\log(-\theta) \\
c(y,\phi) &= 0
\end{aligned}$$

2.1.3 Gamma(β, α) when α is known

$$f(y;\beta,\alpha) = \frac{y^{\alpha-1}\beta^{\alpha}}{\Gamma(\alpha)}\exp(-y\beta)$$

let $\beta = \alpha/\mu$

$$\begin{aligned}
&= (\alpha/\mu)^{\alpha}\frac{y^{\alpha-1}}{\Gamma(\alpha)}\exp(-\alpha y/\mu) \\
&= \exp\left(-\frac{\alpha y}{\mu} + (\alpha-1)\log(y) - \log(\Gamma(\alpha)) + \alpha\log(\alpha) - \alpha\log(\mu)\right) \\
&= \exp\left(\left[\frac{y}{-\mu} - \log(\mu)\right]\alpha + (\alpha-1)\log(y) - \log(\Gamma(\alpha)) + \alpha\log(\alpha)\right)
\end{aligned}$$

where

$$\begin{aligned}
a(\phi) &= \frac{1}{\alpha} \\
\theta &= \frac{1}{-\mu} = -\frac{\beta}{\alpha} \\
b(\theta) &= \log(\mu) = -\log\left(\frac{-1}{\theta}\right) = \log(-\theta) \\
c(y,\phi) &= (\alpha-1)\log(y) - \log(\Gamma(\alpha)) + \alpha\log(\alpha)
\end{aligned}$$

2.1.4 Inverse-Normal (too lazy)

2.1.5 Normal(μ, σ^2) for known σ^2

$$\begin{aligned}
f(y;\mu) &= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) \\
&= \exp\left(-\frac{1}{2\sigma^2}(y^2 - 2y\mu + \mu^2) - \frac{1}{2}\log(\pi\sigma^2)\right) \\
&= \exp\left((y\mu - \frac{\mu^2}{2})/\sigma^2 - \frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log(\pi\sigma^2)\right)\right)
\end{aligned}$$

where

$$\begin{aligned}
a(\phi) &= \sigma^2 \\
\theta &= \mu \\
b(\theta) &= \mu^2/2 = \theta^2/2 \\
c(y,\phi) &= -\frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log(\pi\sigma^2)\right)
\end{aligned}$$

2.1.6 Poisson(μ)

$$\begin{aligned}
f(y;\mu) &= \frac{\mu^y\exp(-\mu)}{y!} \\
&= \exp\left(y\log(\mu) - \mu - \log(y!)\right)
\end{aligned}$$

where

$$
\begin{aligned}
a(\phi) &= 1 \\
\theta &= \log(\mu) \\
b(\theta) &= \mu \\
c(y, \phi) &= -\log(y!)
\end{aligned}
$$

2.2 Mean and Variance Estimation

The parameters θ and ϕ are location and scale parameters, respectively. It can be shown that if y_i follows a distribution in the EF, then it has a <u>mean and variance</u> which can be estimated by

$$E(y_i) = \mu_i = b'(\theta_i) \tag{2}$$

$$Var(y_i) = \sigma_i^2 = b''(\theta_i) \cdot a_i(\phi) \tag{3}$$

where $b'(\theta_i)$ and $b''(\theta_i)$ are the first and second derivatives of $b(\theta_i)$. *note:* we often denote $b''(\theta_i)$ as the <u>Variance Function</u>.

$$b''(\theta_i) = v(\mu_i) \tag{4}$$

The derivations can be done solved for by using the well known relations

$$\text{mean can be solved from} \rightarrow E\left(\frac{\partial l}{\partial \theta}\right) = 0$$

$$\text{variance can be solved from} \rightarrow E\left(\frac{\partial^2 l}{\partial \theta^2}\right) - E\left(\left[\frac{\partial l}{\partial \theta}\right]^2\right) = 0$$

where l is the log-likelihood, $l = \log(L(\theta; y, \phi)) = \log(f(y; \theta, \phi))$.

$$
\begin{aligned}
0 &= E\left(\frac{\partial l(\theta; y, \phi)}{\partial \theta}\right) \\
&= \int_y \frac{\partial l(\theta; y, \phi)}{\partial \theta} f(y; \theta, \phi) dy
\end{aligned}
$$

definition of $\frac{d}{dx}\log(f(x)) = (\frac{d}{dx}f(x))(\frac{1}{f(x)})$

$$
\begin{aligned}
&= \int_y \left[\frac{\partial}{\partial \theta} f(y; \theta, \phi)\right] \frac{1}{f(y; \theta, \phi)} f(y; \theta, \phi) dy \\
&= \int_y \frac{\partial}{\partial \theta} f(y; \theta, \phi) dy
\end{aligned}
$$

definition of $\frac{d}{dx}\exp(f(x)) = (\frac{d}{dx}f(x))(\exp(f(x)))$ and $f(y; \theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$

$$= \int_y \left[\frac{\partial}{\partial \theta}\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)\right] f(y; \theta, \phi) dy$$

I want the reader to note **we could have just skipped directly to the step above with the understanding that** $\log(\exp(x)) = x$; *however*, I wanted to demonstrate that following the order of operations within calculus leads to the same result.

$$
\begin{aligned}
&= \int_y \left[\frac{y}{a(\phi)} - \frac{b'(\theta)}{a(\phi)}\right] f(y;\theta,\phi)dy \\
&= \frac{1}{a(\phi)} \int_y y f(y;\theta,\phi)dy - \frac{b'(\theta)}{a(\phi)} \int_y f(y;\theta,\phi)dy \\
&= \frac{1}{a(\phi)} \cdot E(y) - \frac{b'(\theta)}{a(\phi)} \cdot 1 \\
\implies E(y) &= b'(\theta)
\end{aligned}
$$

Similarly, for variance:

$$
\begin{aligned}
0 &= E\left(\frac{\partial^2 l}{\partial \theta^2}\right) - E\left(\left[\frac{\partial l}{\partial \theta}\right]^2\right) \\
&= E\left(\frac{b''(\theta)}{a(\phi)}\right) - E\left(\frac{(y-b'(\theta))^2}{a^2(\phi)}\right) \\
&= \frac{b''(\theta)}{a(\phi)} - \frac{1}{a^2(\phi)}\left[\int_y \left[y^2 - 2yb'(\theta) + b'(\theta)^2\right] f(y;\theta,\phi)dy\right] \\
&= \frac{b''(\theta)}{a(\phi)} - \frac{1}{a^2(\phi)}\left[\int_y y^2 f(y;\theta,\phi)dy - 2b'(\theta)\int_y y f(y;\theta,\phi)dy + b'(\theta)^2 \int_y f(y;\theta,\phi)dy\right] \\
&= \frac{b''(\theta)}{a(\phi)} - \frac{1}{a^2(\phi)}\left[E(y^2) - 2b'(\theta)E(y) + b'(\theta)^2\right]
\end{aligned}
$$

recall $b'(\theta) = E(y)$

$$
\begin{aligned}
&= \frac{b''(\theta)}{a(\phi)} - \frac{1}{a^2(\phi)}\left[E(y^2) - E(y)^2\right] \\
&= \frac{b''(\theta)}{a(\phi)} - \frac{1}{a^2(\phi)}\text{var}(y) \\
\implies \text{var}(y) &= b''(\theta) \cdot a(\phi)
\end{aligned}
$$

3 Maximum Likelihood Estimation

Let $y = [y_1, \ldots, y_n]^T$ denote a random vector and let the joint p.d.f. of the y_i's be $f(y;\theta)$, where $f(y;\theta)$ depends on the parameter vector $\theta = [\theta_1, \ldots, \theta_p]^T$. The likelihood function $L(\theta; y)$ is the same as the joint p.d.f. but with a change in notation. In the likelihood function our focus is on θ while we assume y is fixed. The <u>likelihood estimation of θ</u> is the value of θ which maximizes the likelihood function.

$$L(\hat{\theta}; y) > L(\theta; y) \; \forall \theta \neq \hat{\theta}$$

Since the logarithm function is monotonically increasing, it follows that $\log(L(\theta; y)) = l(\theta; y)$ will be maximized at the same point, that being $\hat{\theta}$. The <u>maximum likelihood estimator, $\hat{\theta}$,</u> is obtained by solving the simultaneous equations.

$$\frac{\partial l(\theta; y)}{\partial \theta_j} = 0 \; \forall j = 1, \ldots, p$$

3.1 Properties of MLE

(i) The MLE is asymptotically distributed as

$$\hat{\theta} \sim N(\theta, I^{-1})$$

where I is the information matrix whose elements are

$$I_{ij} = -E\left(\frac{\partial^2 l(\theta; y)}{\partial\theta_i \partial\theta_j}\right)$$

(ii) *The invariance property of MLE's:* If $\hat{\theta}$ is the MLE and $g(\theta)$ is any function of θ, then the MLE of $g(\theta)$ is $g(\hat{\theta})$.

(iii) Define

$$u_j = \frac{\partial l(\theta; y)}{\partial\theta_j} \text{ where } j = 1, \ldots, p$$

The <u>Score Vector</u> is at least asymptotically distributed as

$$U = \begin{bmatrix} u_1 \\ \vdots \\ u_p \end{bmatrix} \sim N(0, I)$$

Therefore, for large samples,

$$U^T I^{-1} U \sim \chi^2_{(p)}$$

3.2 Finding the MLE of β within a GLM

Within a general linear model we have a specific statistic that is being estimated via regression through the link function g. This link function was defined above as

$$E(y_i) = g(\mu_i) = X_i^T \beta$$

A trick to get a proper estimation of β is use another variable to link the regression to the link function, we do this normally through the following set of relations.

1. $\eta_i = X_i^T \beta$
2. $g(\mu_i) = \eta_i$
3. $\mu_i = b'(\theta_i)$
4. $v(\mu_i) = b''(\theta_i)$

Let's see how these relations allow us to get an estimate for a single β_j. We do this by calculating the score function for a single β_j.

$$\frac{\partial l}{\partial \beta_j} = \sum_{i=1}^{n} \left[\frac{\partial l}{\partial \theta_i}\right] \cdot \left[\frac{\partial \theta_i}{\partial \mu_i}\right] \cdot \left[\frac{\partial \mu_i}{\partial \eta_i}\right] \cdot \left[\frac{\partial \eta_i}{\partial \beta_j}\right]$$

Here we just expand over i.

$$
\begin{aligned}
&= \sum_{i=1}^{n} \left[\frac{y_i - b'(\theta_i)}{a(\phi_i)}\right] \cdot \left[\frac{\partial \mu_i}{\partial \theta_i}\right]^{-1} \cdot \left[\frac{\partial \eta_i}{\partial \mu_i}\right]^{-1} \cdot \left[\frac{\partial \eta_i}{\partial \beta_j}\right] \\
&= \sum_{i=1}^{n} \left[\frac{y_i - b'(\theta_i)}{a(\phi_i)}\right] \cdot \left[\frac{\partial}{\partial \theta_i} b'(\theta_i)\right]^{-1} \cdot \left[\frac{\partial}{\partial \mu_i} g(\mu_i)\right]^{-1} \cdot \left[\frac{\partial}{\partial \beta_j} X_i^T \beta\right] \\
&= \sum_{i=1}^{n} \frac{y_i - b'(\theta_i)}{a(\phi_i)} \cdot \frac{1}{b''(\theta_i)} \cdot \frac{1}{g'(\mu_i)} \cdot x_{ij} \\
&= \sum_{i=1}^{n} \frac{y_i - \mu_i}{[b''(\theta_i) \cdot a(\phi_i)]} \cdot \frac{1}{g'(\mu_i)} \cdot x_{ij}
\end{aligned}
$$

So the score for the jth term of β is

$$
U_j = \sum_{i=1}^{n} \frac{y_i - \mu_i}{\text{var}(y_i) g'(\mu_i)} x_{ij}
$$

and in matrix form we have

$$
\frac{\partial l}{\partial \beta} = U = X^T W \Delta (y_i - \mu_i)
$$

where

$$
\begin{aligned}
W &= diag\left(\frac{1}{a(\phi_i) v(\mu_i)} \cdot \frac{1}{g'(\mu_i)^2}\right) \\
\Delta &= diag(g'(\mu_i))
\end{aligned}
$$

Note the algebra to get here is not too complicated, see that

$$
\begin{aligned}
U_j &= \sum_{i=1}^{n} \frac{y_i - \mu_i}{\text{var}(y_i) g'(\mu_i)} x_{ij} \\
&= \sum_{i=1}^{n} \frac{y_i - \mu_i}{b''(\theta_i) \cdot a(\phi_i) \cdot g'(\mu_i)} \frac{g'(\mu_i)}{g'(\mu_i)} x_{ij} \\
&= \sum_{i=1}^{n} (y_i - \mu_i) \frac{1}{v(\mu_i) a(\phi_i) \cdot g'(\mu_i)^2} \frac{g'(\mu_i)}{1} x_{ij}
\end{aligned}
$$

Next let's find the hessian and expected hessian.

$$
\begin{aligned}
\frac{\partial^2 l}{\partial\beta_j\partial\beta_j'} &= \frac{\partial}{\partial\beta_j'}\sum_{i=1}^{n}\frac{y_i-\mu_i}{a(\phi_i)v(\mu_i)g'(\mu_i)}x_{ij} \\
&= \frac{1}{a(\phi_i)}\sum_{i=1}^{n}\frac{\partial}{\partial\beta_j'}\left[\frac{y_i-\mu_i}{v(\mu_i)g'(\mu_i)}\right]x_{ij} \\
&= \frac{1}{a(\phi_i)}\sum_{i=1}^{n}\left[\frac{\partial}{\partial\beta_j'}\{y_i-\mu_i\}\frac{1}{v(\mu_i)g'(\mu_i)}+(y_i-\mu_i)\frac{\partial}{\partial\beta_j'}\left\{\frac{1}{v(\mu_i)g'(\mu_i)}\right\}\right]x_{ij} \\
&= \frac{1}{a(\phi_i)}\sum_{i=1}^{n}\left[\frac{\partial}{\partial\mu_i}\frac{\partial\mu_i}{\partial\eta_i}\frac{\partial\eta_i}{\partial\beta_j'}\{y_i-\mu_i\}\frac{1}{v(\mu_i)g'(\mu_i)}+(y_i-\mu_i)\frac{\partial}{\partial\beta_j'}\left\{\frac{1}{v(\mu_i)g'(\mu_i)}\right\}\right]x_{ij} \\
&= \frac{1}{a(\phi_i)}\sum_{i=1}^{n}\left[\frac{1}{g'(\mu_i)}x_{ij}'\{-1\}\frac{1}{v(\mu_i)g'(\mu_i)}+(y_i-\mu_i)\frac{\partial}{\partial\mu_i}\frac{\partial\mu_i}{\partial\eta_i}\frac{\partial\eta_i}{\partial\beta_j'}\left\{\frac{1}{v(\mu_i)g'(\mu_i)}\right\}\right]x_{ij} \\
&= \frac{1}{a(\phi_i)}\sum_{i=1}^{n}\frac{1}{g'(\mu_i)}x_{ij}'\left[\frac{-1}{v(\mu_i)g'(\mu_i)}+(y_i-\mu_i)\frac{\partial}{\partial\mu_i}\left\{\frac{1}{v(\mu_i)g'(\mu_i)}\right\}\right]x_{ij} \\
&= \frac{-1}{a(\phi_i)}\sum_{i=1}^{n}\frac{1}{g'(\mu_i)}x_{ij}'\left[\frac{1}{v(\mu_i)g'(\mu_i)}+(y_i-\mu_i)\left\{\frac{v'(\mu_i)g'(\mu_i)+v(\mu_i)g''(\mu_i)}{(v(\mu_i)g'(\mu_i))^2}\right\}\right]x_{ij} \\
E\left[\frac{-\partial^2 l}{\partial\beta_j\partial\beta_j'}\right] &= \frac{-1}{a(\phi_i)}\sum_{i=1}^{n}\frac{1}{v(\mu_i)g'(\mu_i)^2}x_{ij}'x_{ij} \\
&= X_j^T W X_j
\end{aligned}
$$

This is called the <u>Information</u> based on the fisher scoring method for a single β_j. We can use Newton Raphson to get an estimation of it. When we implore this model for the full coefficient vector where we label it as $\hat{V}_H$ and denote it the "expected Hessian". It is defined as

$$\hat{V}_H = \left\{-E\left[\frac{\partial^2 l}{\partial\beta\partial\beta^T}\right]\right\}^{-1}.$$

A more robust approach to this variance estimator was found by Huber (1967) with a sandwich approach of the following form

$$\hat{V}_S = \hat{V}_H \cdot \hat{B}_S \cdot \hat{V}_H$$

where

$$\hat{B}_S = \sum_{i=1}^{n} x_i\left[\frac{(y_i-\hat{\mu}_i)}{a(\phi)v(\mu_i)g'(\mu_i)}\right]^2 x_i^T$$

and

$$\hat{V}_H = \left\{-\sum_{i=1}^{n} x_i\left[\frac{1}{a(\phi_i)v(\mu_i)g'(\mu_i)^2}\right]x_i^T\right\}^{-1} = \left\{X^T W X\right\}^{-1}$$

4 Model Estimation

4.1 Methods for finding MLE for β

The Newton-Raphson Method for finding an estimate of θ_j was derived by using the Taylor expansion on $\frac{\partial l(\theta;y)}{\partial \theta_j}$ around $\theta = \theta^{(0)}$ the follows yields,

$$\frac{\partial l(\theta;y)}{\partial \theta_j} = \frac{\partial l(\theta;y)}{\partial \theta_j}\bigg|_{\theta=\theta^{(0)}} + \frac{\partial^2 l(\theta;y)}{\partial \theta_j \partial \theta_j}\bigg|_{\theta=\theta^{(0)}} (\theta_j - \theta_j^{(0)}) + O(n^2)$$

Solving for θ then gives,

$$\theta_j = \theta_j^{(0)} - \left(\left[\frac{\partial^2 l(\theta;y)}{\partial \theta_j \partial \theta_j}\bigg|_{\theta=\theta^{(0)}} \right]^{-1} \cdot \frac{\partial l(\theta;y)}{\partial \theta_j}\bigg|_{\theta=\theta^{(0)}} \right)$$

This iterative result is the Newton-Raphson method. The same method can be applied to solve for all β values. For example,

$$\beta = \hat{\beta} - \left(\left[\frac{\partial^2 l}{\partial \beta \partial \beta^T} \right]^{-1} \cdot \frac{\partial l}{\partial \beta} \right)_{\beta=\hat{\beta}} \tag{5}$$

A common starting location for this algorithm is to assume an intercept and all other β values to be zero.

4.2 Fisher Scoring Method

The fisher scoring method is the same iteration method as the Newton Raphson but with a slight change, we replace the 2nd derivative of $l(\theta;y)$ by it's expectation.

$$\theta = \theta^{(0)} - \left(\left[E\left\{ \frac{\partial^2 l(\theta;y)}{\partial \theta \partial \theta^T} \right\}\bigg|_{\theta=\theta^{(0)}} \right]^{-1} \cdot \frac{\partial l(\theta;y)}{\partial \theta}\bigg|_{\theta=\theta^{(0)}} \right)$$

We may also apply this to β just as we did before.

$$\beta = \beta^{(0)} - \left(\left[E\left\{ \frac{\partial^2 l(\beta;y)}{\partial \beta \partial \beta^T} \right\}\bigg|_{\beta=\beta^{(0)}} \right]^{-1} \cdot \frac{\partial l(\beta;y)}{\partial \beta}\bigg|_{\beta=\beta^{(0)}} \right) \tag{6}$$

4.3 IRLS

This is <u>IRLS</u> method or <u>Iterative Re-weighted Least Squares</u> method is based on the fisher scoring method. The method is an iterative method that continues moving the values of the coefficients closer to the relative maximum near the initial location. The derivation is as follows:

$$\beta = \hat{\beta} - \left(E\left[\frac{\partial^2 l}{\partial \beta \partial \beta^T} \right]^{-1} \cdot \frac{\partial l}{\partial \beta} \right)_{\beta=\hat{\beta}}$$

$$-E\left[\frac{\partial^2 l}{\partial \beta \partial \beta^T} \right]_{\beta=\hat{\beta}} \cdot \beta = -E\left[\frac{\partial^2 l}{\partial \beta \partial \beta^T} \right]_{\beta=\hat{\beta}} \cdot \hat{\beta} + \left(\frac{\partial l}{\partial \beta} \right)_{\beta=\hat{\beta}}.$$

We then replace the information and score with their estimates. Recall:

$$U = \frac{\partial l}{\partial \beta} = X^T W \Delta(y_i - \mu_i)$$
$$I = E\left[\frac{-\partial^2 l}{\partial \beta \partial \beta^T}\right] = X^T W X$$

So we have,

$$\begin{aligned} X^T W^{(r-1)} X \beta^{(r)} &= X^T W^{(r-1)} X \beta^{(r-1)} + X^T W^{(r-1)} \Delta(y - \mu^{(r-1)}) \\ &= X^T W^{(r-1)} \underbrace{\left\{X\beta^{(r-1)} + \Delta(y - \mu^{(r-1)})\right\}}_{Z^{(r-1)}} \end{aligned}$$

where the $(r-1)$ superscript denotes the initial value of the β vector, and (r) represents the new value of the beta vector being solved for. More specifically

$$\beta^{(r)} = \left[\{X^T W^{(r-1)} X\}^{-1}\right]\left[X^T W^{(r-1)} Z^{(r-1)}\right]$$

with

$$\begin{aligned} Z_i^{(r-1)} &= X_i \beta^{(r-1)} + \Delta_i (y_i - \mu_i^{(r-1)}) \\ &= (y_i - \mu_i^{(r-1)}) g'(\mu_i^{(r-1)}) + (\eta_i^{(r-1)} - \text{offset}) \end{aligned}$$

where the offset is model dependent. We choose the starting value of μ_i to be $\frac{y_i + \bar{y}}{2}$ for non-binomial models, and $\frac{r_i(y+0.5)}{r_i+1}$ for $y \sim Binomial(r_i, p_i)$

4.4 Bringing it together

We have a distribution function that belongs to the exponential family; it can be represented by the equation

$$f(y; \theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right) \tag{7}$$

where θ is the parameter of interest and ϕ is known (nuisance). Distributions of this form have mean and variance

$$E(y) = b'(\theta)$$
$$\text{var}(y) = b''(\theta) \cdot a(\phi)$$

We wish to estimate the parameter θ using a list of predictors X and coefficients β with the relations

1. $\eta_i = X_i^T \beta$
2. $g(\mu_i) = \eta_i$
3. $\mu_i = b'(\theta_i)$
4. $v(\mu_i) = b''(\theta_i)$

The relations give us the score and information functions

$$U = \frac{\partial l}{\partial \beta} = X^T W \Delta(\boldsymbol{y} - \boldsymbol{\mu})$$
$$I = E\left[\frac{-\partial^2 l}{\partial \beta \partial \beta^T}\right] = X^T W X$$

where

$$W = diag\left(\frac{1}{a(\phi_i)v(\mu_i)} \cdot \frac{1}{g'(\mu_i)^2}\right)$$
$$\Delta = diag(g'(\mu_i)).$$

From here we can begin to look at the estimation procedures used to solve for the coefficients represented by the score (estimating function) and information (hessian). From here we use the Taylor expansion on the score function to solve for the regression coefficients, β.

$$\beta = \hat{\beta} - \left[I^{-1} \cdot U\right]_{\beta=\hat{\beta}}$$

With a little bit of manipulation we can create an iterative solving equation to find the values of β which locally maximize the likelihood function.

$$\beta^{(r)} = \left[\{X^T W^{(r-1)} X\}^{-1}\right]\left[X^T W^{(r-1)} Z^{(r-1)}\right]$$

with

$$\begin{aligned} Z_i^{(r-1)} &= X_i \beta^{(r-1)} + \Delta_i (y_i - \mu_i^{(r-1)}) \\ &= (y_i - \mu_i^{(r-1)}) g'(\mu_i^{(r-1)}) + (\eta_i^{(r-1)} - \text{offset}) \end{aligned}$$

where the $(r-1)$ superscript denotes the initial value of the β vector, and (r) represents the new value of the beta vector being solved for, and the offset is model dependent.

5 Model Diagnostics

Before we continue I'd like to quickly reference the asymptotic distributions we are describing with these estimations. In the case of normal errors and a identity link function, we are finding the estimate $\hat{\beta}$ where

$$\hat{\beta} \xrightarrow{d} \mathrm{N}_p(\beta, \sigma^2 (X^T X)^{-1})$$

When we change the function from the link function to another (and maintain normal error), we then find that we have the asymptotic distribution of

$$\hat{\beta} \xrightarrow{d} \mathrm{N}_p(\beta, (X^T W X)^{-1})$$

where

$$W = diag\left(\frac{1}{a(\phi_i)v(\mu_i)} \cdot \frac{1}{g'(\mu_i)^2}\right)$$

5.1 Hypothesis Testing

Tests for β are based on the corresponding marginal moment distribution. For example, in the case of normal errors and identity link function we would use the same tests as a regular regression on the distribution

$$\hat{\beta} \xrightarrow{d} \mathrm{N}_p(\beta, \sigma^2(X^T X)^{-1}).$$

5.1.1 Likelihood Ratio Test & Deviance

The likelihood ratio test is of the form

$$\frac{L(\hat{\theta}; y, \phi)}{L(\tilde{\theta}; y, \phi)}$$

where $\hat{\theta}$ represents the estimated values of the parameters under the partial model (or null hypothesis), and $\tilde{\theta}$ represents the value of the parameter under the full model (or alternative hypothesis). When we use the log-likelihood we end up with the formula that resembles the following form:

$$\frac{y\hat{\theta} - b(\hat{\theta})}{a(\phi)} + c(y, \phi) - \left[\frac{y\tilde{\theta} - b(\tilde{\theta})}{a(\phi)} + c(y, \phi)\right] = \frac{y(\hat{\theta} - \tilde{\theta}) - (b(\hat{\theta}) - b(\tilde{\theta}))}{a(\phi)}.$$

The likelihood criterion to compare these two models in the exponential family has the form

$$-2\log(\lambda) = 2\sum_{i=1}^{n} \frac{y_i(\tilde{\theta}_i - \hat{\theta}_i) - (b(\tilde{\theta}_i) - b(\hat{\theta}_i))}{a(\phi)}$$

where $\lambda = \frac{L(\hat{\theta}; y, \phi)}{L(\tilde{\theta}; y, \phi)}$. The expression on the right-hand-side above is called *deviance*.

$$-2\log(\lambda) = D(y, \hat{\theta})$$

5.1.2 Pearson Residuals

Residuals are also useful tools for checking the assumptions made in formulating a model. Suppose $E(Y_i) = \mu_i$ and $Y_i \sim N(\mu, \sigma^2)$, the approximate residuals are defined by,

$$r_i^p = \frac{y_i - \hat{\mu}_i}{\hat{\sigma}} = \frac{y_i - \hat{\mu}_i}{\sqrt{\sigma^2(1 - h_{ii})}}$$

where h_{ii} are the ii^{th} entries of the Hessian matrix $(X(X^T X)^{-1} X^T)$.

$$\sum (r_i^p)^2 \sim \chi^2_{(n-p)}$$

5.1.3 Deviance Residuals

$$\begin{aligned} r_i^d &= \text{sign}(y_i - \hat{\mu}_i)\sqrt{d} \\ d &= \text{sign}(o_i - e_i)\sqrt{2[o_i \log(o_i/e_i) - (o_i - e_i)]} \\ D = \sum d_i^2 &\sim \chi^2_{(n-p)} \end{aligned}$$

5.2 R Code for Newton Raphson Method:

```
x=rep(1:8)
n=c(17742,16554,16059,13083,10784,9645,10706,9933)
y=c(1,5,5,12,25,38,54,65)
X=cbind(rep(1, length(x)), x)
ratio=y/n

plot(x, log(ratio))
lm1=summary(lm(log(ratio)~x))
lines(x, lm1$coefficients[1] + lm1$coefficients[2]*x)

likelihood=function(y, x, n, b){
  temp=0
  for (i in 1:length(y)) {
    temp=temp+y[i]*(b[1]+b[2]*x[i])-n[i]*exp(b[1]+b[2]*x[i])
              +y[i]*log(n[i])-log(factorial(y[i]))
  }
  return(temp)
}

first.der=function(y, x, n, b){
  b1 = c(0,0)
  for (i in 1:length(y)) {
    b1[1] = b1[1]+y[i]-n[i]*exp(b[1]+b[2]*x[i])
    b1[2] = b1[2]+y[i]*x[i]-n[i]*x[i]*exp(b[1]+b[2]*x[i])
  }
  return(b1)
}

second.der=function(y, x, n, b){
  mat=matrix(0,2,2)
  for (i in 1:length(y)) {
    mat[1,1]=mat[1,1]-n[i]*exp(b[1]+b[2]*x[i])
    mat[1,2]=mat[1,2]-n[i]*x[i]*exp(b[1]+b[2]*x[i])
    mat[2,2]=mat[2,2]-n[i]*x[i]*x[i]*exp(b[1]+b[2]*x[i])
  }
  mat[2,1]=mat[1,2]
}

#Newton Raphson Method
sec.d=matrix(0,2,2)
beta.new=c(lm1$coefficients[1],lm1$coefficients[2])
fir.d <- beta.old <- c(0,0)
damp=0.1

while(max(abs(beta.old-beta.new))>0.001){
  beta.old=beta.new
  fir.d=first.der(y, x, n, beta.old)
  sec.d=second.der(y, x, n, beta.old)
  temp=solve(sec.d)
  beta.new=beta.old - temp%*%fir.d*damp
  print(beta.new)
}
```

```
lines(x, beta.new[1] + beta.new[2]*x, col=2)
```

5.3 R Code for IRLS

```
x= c(1:8)
y= c(1, 5, 5, 12, 25, 38, 54, 65)
n= c(17742, 16554, 16059, 13083, 10784, 9645, 10704, 9933)
X= cbind(rep(1,length(y)), x, x^2)

mu0 = (y + mean(y))/2
w0 = diag(mu0)
X1 = t(X)%*%w0%*%X
##z=xb+delta(y-u)= log(u0) - log(u)   +   delta(y-u)
##                 ^^^nu-offset^^^      ^^^(y-u)g'(u)^^^
z0 = log(mu0)-log(n) + (y-mu0)/mu0
beta0 = solve(X1)%*%t(X)%*%w0%*%z0 #beta1 in notes

mu0 = as.vector(n)*exp(X%*%beta0)
w0 = diag(as.vector(mu0))
X1 = t(X)%*%w0%*%X
z0 = log(mu0) - log(n) + (y-mu0)/(mu0)
beta1 = solve(X1)%*%t(X)%*%w0%*%z0

count=0
while (max(abs(beta1-beta0))>0.00001) {
  beta0 = beta1
  mu0 = as.vector(n)*exp(X%*%beta0)
  w0 = diag(as.vector(mu0))
  X1 = t(X)%*%w0%*%X
  z0 = log(mu0) - log(n) + (y-mu0)/(mu0)
  beta1 = solve(X1)%*%t(X)%*%w0%*%z0
  count = count + 1
}

variance = solve(X1)
print(beta1)
print(count)
print(variance)
```

5.4 R Code for Defining your own GLM

```
#GLM link define example
dummy.name <- function(){
  ## link
  linkfun <- function(y){ -log(-log(y))}
  ## inverse link
  linkinv <- function(eta){ exp(-exp(-eta))}
  ## derivative of invlink wrt eta
  mu.eta <- function(eta){  exp(-eta-exp(-eta))}
  valideta <- function(eta) TRUE
  link <- "loglog"
  structure(list(linkfun = linkfun, linkinv = linkinv,
                 mu.eta = mu.eta, valideta = valideta,
                 name = link),
            class = "link-glm")
```

```
}

loglog <- dummy.name()

#example
x = c(1.6907, 1.7242, 1.7552, 1.7842, 1.8113, 1.8369, 1.8610, 1.8839)
X = cbind(rep(1, length(x)), x)
n = c(59, 60, 62, 56, 63, 59, 62, 60)
y = c(6, 13,18,28,52,53,61,60)
ratio = y/n

glm(ratio ~ x, family = binomial(link=loglog))
```